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Lehrbuch der Algebra
Working English Translation Batch 11
Galois Groups, Metacyclic Groups, and Abstract Groups

Translated into modern mathematical English

Translator's note

This batch continues Weber's transition from invariant theory and linear substitutions to Galois groups and the abstract group concept.

§44. Galois groups and resolvents

The theory of groups of linear substitutions is closely related to the theory of algebraic equations. Let

$$t^n + a_1 t^{n-1} + a_2 t^{n-2} + \cdots + a_n = 0 \tag{1}$$

be an equation of degree n , with roots

$$\alpha_1, \alpha_2, \dots, \alpha_n.$$

The totality of all permutations of these roots which leave unchanged every rational relation among them forms the *Galois group* of the equation.

A function

$$F(\alpha_1, \dots, \alpha_n)$$

of the roots that is unchanged by all permutations of the Galois group is rationally expressible through the coefficients

$$a_1, \dots, a_n$$

of the equation.

Solving the equation is equivalent to determining a function ϑ of the roots that is left fixed by no nonidentity permutation of the Galois group. Such a function is called a *primitive function* of the field.

The Galois group may always be regarded as a group of linear substitutions: replace each permutation of the roots by the corresponding substitution of the variables

$$x_1, \dots, x_n,$$

as described in §39. In this way the group of the equation becomes a concrete linear substitution group, and the methods of invariant theory become available for the study of algebraic equations.

Resolvents arise by passing to subgroups. If a function of the roots is invariant under a subgroup S' but not under the whole group S , then the conjugates of that function under S are indexed

by the cosets of S' . Their elementary symmetric functions are invariant under S , hence rational functions of the coefficients of the original equation. The function therefore satisfies an auxiliary equation whose degree is the index $[S : S']$.

§45. Metacyclic groups

A group S is called *metacyclic* if it has a distinguished chain of subgroups

$$S, S_1, S_2, \dots, S_\nu = 1, \quad (2)$$

such that each S_i is a normal subgroup of S_{i-1} and each quotient group

$$S_{i-1}/S_i$$

is cyclic.

Such a group can be built by solving a chain of equations, each of which is cyclic. The solvability of an algebraic equation by radicals is therefore expressed group-theoretically by the metacyclic nature of its Galois group, or equivalently by the existence of a suitable chain with cyclic factors.

For the symmetric group on n letters, metacyclicity occurs only for $n \leq 4$. Likewise the alternating group on n letters is metacyclic only for $n \leq 4$. This is the group-theoretic obstruction behind the impossibility, in general, of solving equations of degree at least five by radicals.

§46. Abstract and concrete groups

Until now the exposition of group theory has been tied to definite operations: substitutions, linear transformations, and similar concrete processes. But the group concept may also be detached from any particular representation and defined purely formally.

By an *abstract group* one understands a system of elements

$$A, B, C, \dots$$

for which a composition law, or multiplication, is defined and satisfies the following rules.

1. From any two elements A, B , their product is a uniquely determined element $C = AB$.
2. The associative law holds:

$$(AB)C = A(BC).$$

3. There is an *identity element* J such that

$$JA = AJ = A$$

for every element A .

4. For every element A there is an inverse element A^{-1} such that

$$AA^{-1} = A^{-1}A = J.$$

Two abstract groups are called *isomorphic* if their elements can be named in such a way that the multiplication rule is the same in both groups. This definition separates the intrinsic structure of the group from the particular objects on which the group happens to act.

The point of abstraction is not to discard concrete examples, but to recognize when two different systems of operations have the same internal law of composition. Permutation groups, linear substitution groups, rotation groups, and Galois groups can then be compared by their multiplication tables rather than by their external form.